14. LOG ANALYSIS FOR DETECTION AND RESPONSE

DATE: 22/04/2026

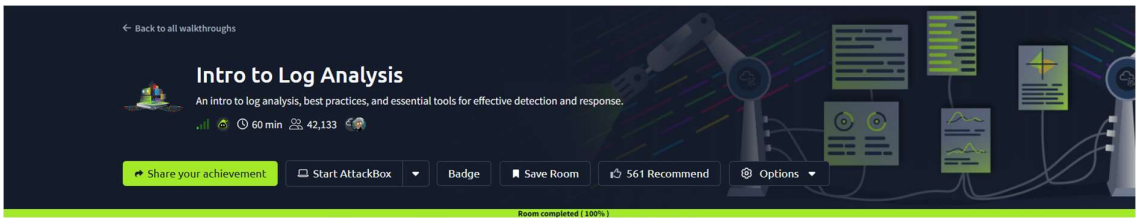
Aim:

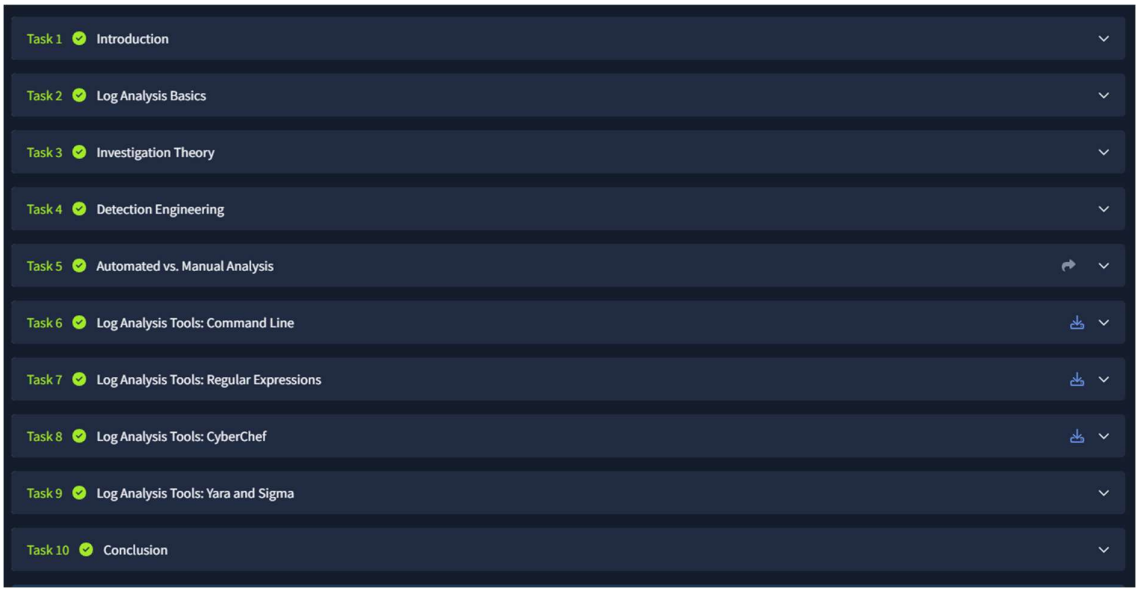
To understand the fundamentals of log analysis and its role in effective security detection and response.

Procedure:

1. Log into TryHackMe and open the Intro to Log Analysis room.
2. Study the fundamentals of log analysis and its role in detection and response.
3. Understand different log types such as system, network, and application logs.
4. Learn best practices for collecting, storing, and analyzing logs.
5. Explore essential tools used for monitoring and identifying suspicious activity.
6. Complete the tasks and answer the questions in the room.
7. Review the summary and confirm completion.

Tasks:





TASK 1 - Introduction

TASK 2 - Log Analysis Basics

Answer the questions below

I understand the basics of logs and I'm ready to proceed!

No answer needed ✓ Correct Answer

TASK 3 - Investigation Theory

Answer the questions below

What's the term for a consolidated chronological view of logged events from diverse sources, often used in log analysis and digital forensics?

Super Timeline ✓ Correct Answer

Which threat intelligence indicator would `5b31f93c09ad1d065c0491b764d04933` and `763f8bdbc98d105a8e82f36157e98bbe` be classified as?

File Hashes ✓ Correct Answer

TASK 4 - Detection Engineering

Answer the questions below

What is the default file path to view logs regarding HTTP requests on an Nginx server?

/var/log/nginx/access.log ✓ Correct Answer

A log entry containing `%2E%2E%2F%2E%2E%2Fproc%2Fself%2Fenviron` was identified. What kind of attack might this infer?

Path Traversal ✓ Correct Answer

TASK 5 – Automated vs Manual Analysis

Answer the questions below

A log file is processed by a tool which returns an output. What form of analysis is this?

Automated ✓ Correct Answer

An analyst opens a log file and searches for events. What form of analysis is this?

Manual ✓ Correct Answer

TASK 6 – Log Analysis Tools:Command Line

```
┌──(JackSparrow㉿Captain)-[~/Desktop]
└─$ cut -d' ' -f7 apache-1691435735822.log | sort |
 uniq
/about.php
/admin.php
/contact.php
/index.php
/index.php?flag=c701d43cc5a3acb9b5b04db7f1be94f6
/login.php
```

```
┌──(JackSparrow㉿Captain)-[~/Desktop]
└─$ grep ' 200 ' apache-1691435735822.log | wc -l
52
```

```
┌──(JackSparrow㉿Captain)-[~/Desktop]
└─$ awk '{print $1}' apache-1691435735822.log | so
rt | uniq -c | sort -nr | head -1
      8 145.76.33.201
```

```
┌──(JackSparrow㉿Captain)-[~/Desktop]
└─$ grep '110.122.65.76' apache-1691435735822.log
| grep '/login.php'
110.122.65.76 - - [31/Jul/2023:12:34:40 +0000] "GE
T /login.php HTTP/1.1" 200 9876 "Mozilla/5.0 (Wind
ows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML
, like Gecko) Chrome/95.0.4638.110 Safari/537.36"
```

Answer the questions below

Use `cut` on the `apache.log` file to return only the URLs. What is the flag that is returned in one of the unique entries?

c701d43cc5a3acb9b5b04db7f1be94f6 — ✓ Correct Answer

In the `apache.log` file, how many total HTTP 200 responses were logged?

52 — ✓ Correct Answer

In the `apache.log` file, which IP address generated the most traffic?

145.76.33.201 — ✓ Correct Answer

What is the complete timestamp of the entry where `110.122.65.76` accessed `/login.php` ?

31/Jul/2023:12:34:40 +0000 — ✓ Correct Answer

TASK 7 – Log Analysis Tools : Regular Expressions

Answer the questions below

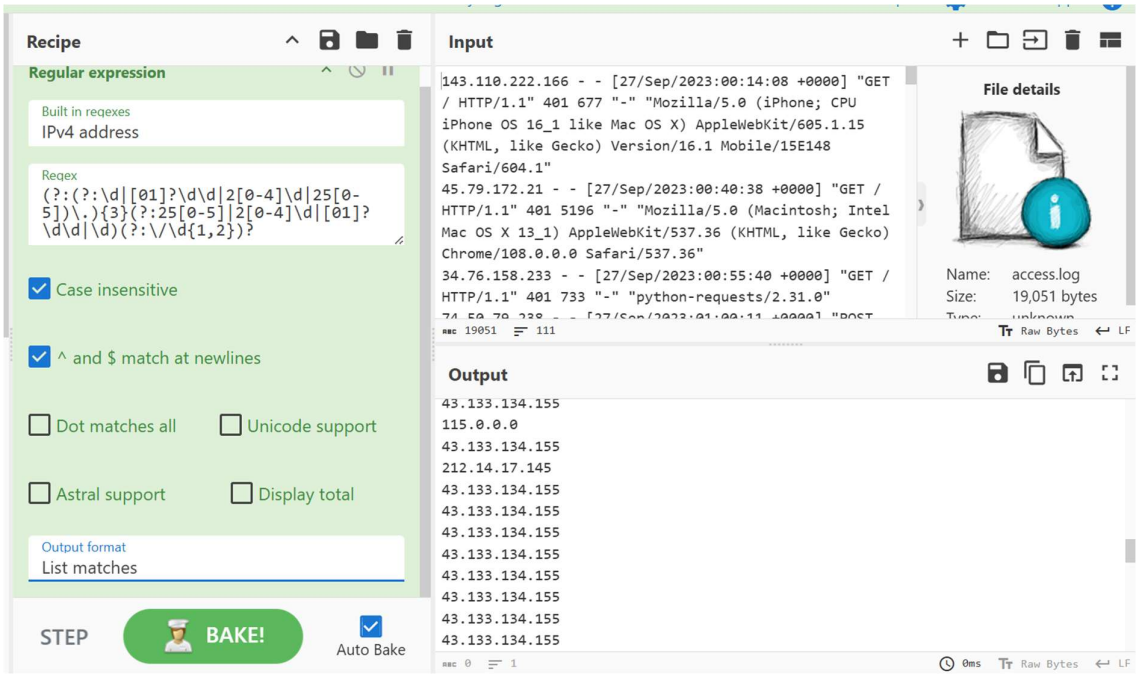
How would you modify the original `grep` pattern above to match blog posts with an ID between 20-29?

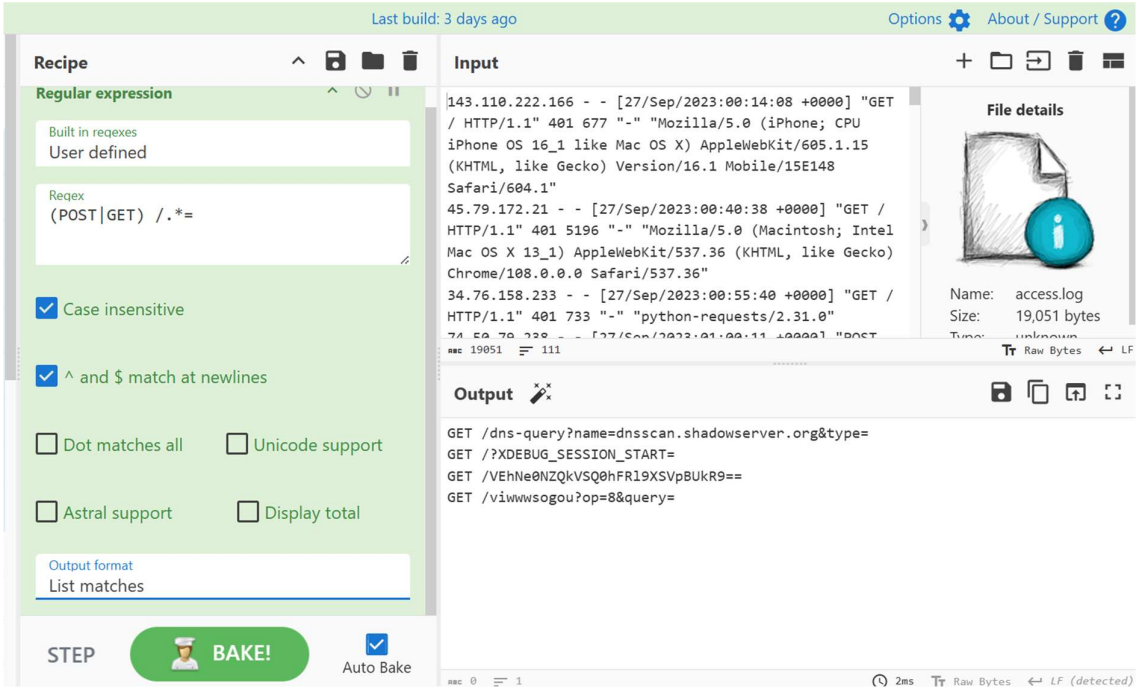
post=2[0-9] — ✓ Correct Answer

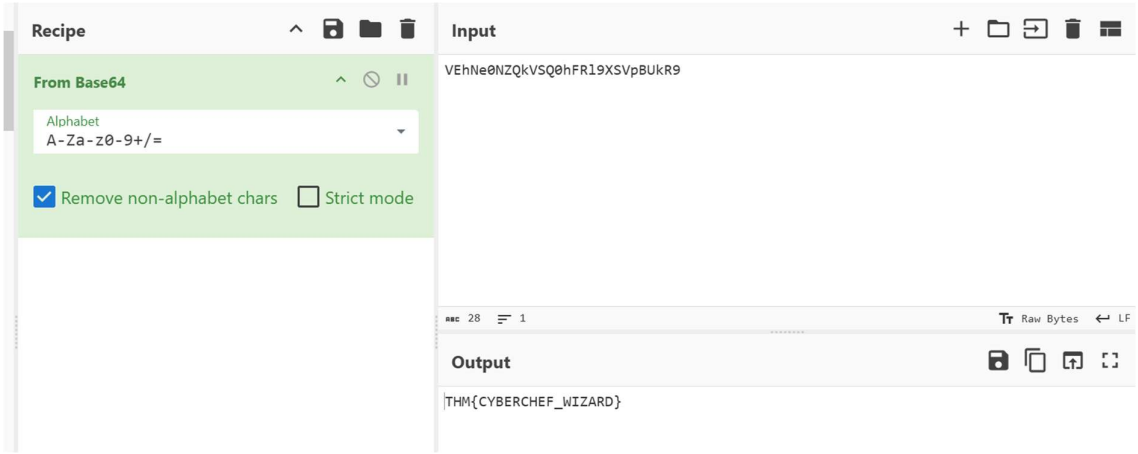
What is the name of the filter plugin used in Logstash to parse unstructured log data?

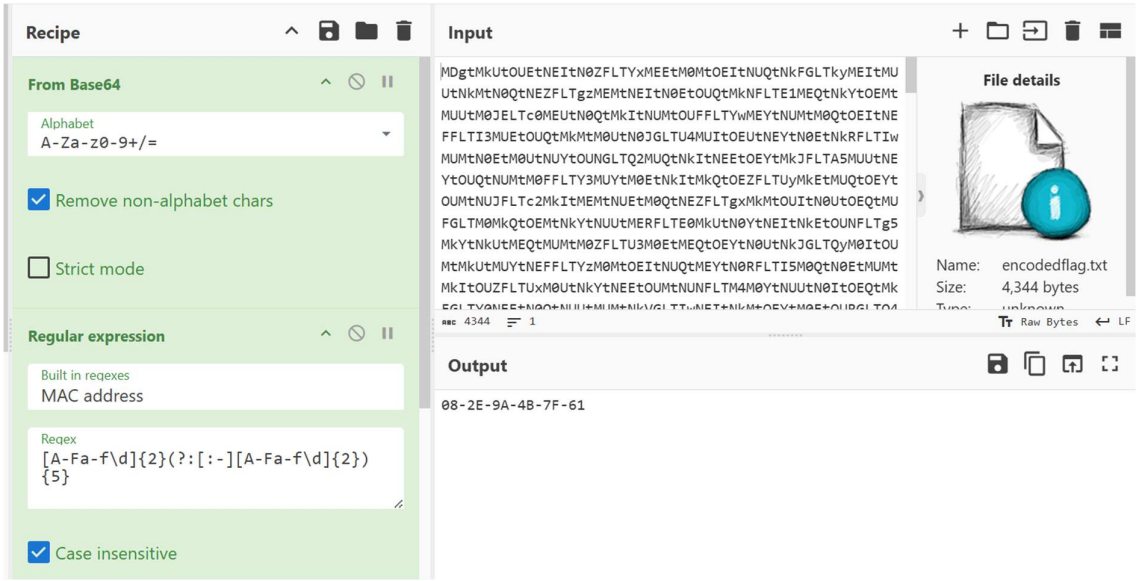
Grok — ✓ Correct Answer

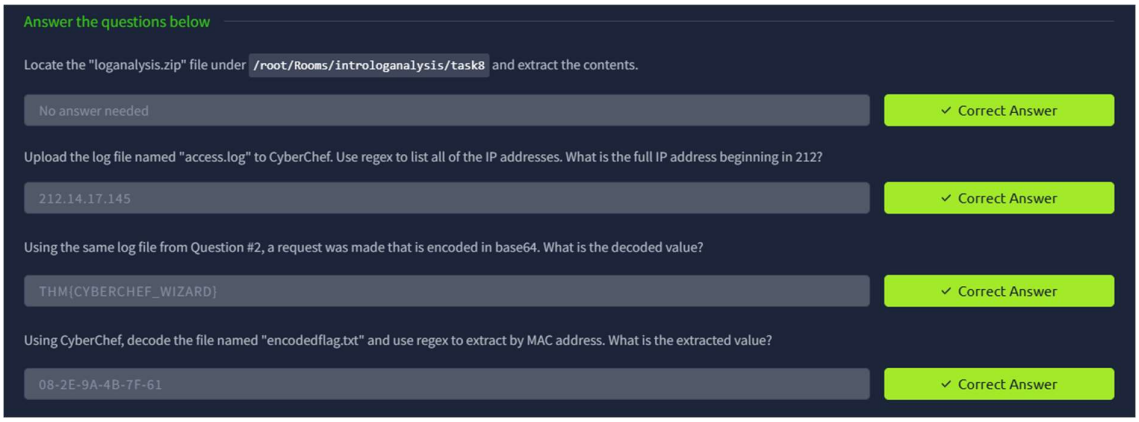
TASK 8 – Log Analysis Tools : CyberChef






Recipe
From Base64
Alphabet
A-Za-z0-9+/=
Remove non-alphabet chars
Strict mode
Input
VEhNe0NZQkVSQ0hFRl9XSVpBUkR9
28 1
Raw Bytes LF
Output
THM{CYBERCHEF_WIZARD}


Recipe
From Base64
Alphabet
A-Za-z0-9+/=
Remove non-alphabet chars
Strict mode
Regular expression
Built in regexes
MAC address
Regex
[A-Fa-f\d]{2}(?:[:-][A-Fa-f\d]{2}){5}
Case insensitive
Input
File details
Name: encodedflag.txt
Size: 4,344 bytes
4344 1
Raw Bytes LF
Output
08-2E-9A-4B-7F-61


Answer the questions below
Locate the "loganalysis.zip" file under /root/Rooms/introloganalysis/task8 and extract the contents.
No answer needed
Correct Answer
Upload the log file named "access.log" to CyberChef. Use regex to list all of the IP addresses. What is the full IP address beginning in 212?
212.14.17.145
Correct Answer
Using the same log file from Question #2, a request was made that is encoded in base64. What is the decoded value?
THM{CYBERCHEF_WIZARD}
Correct Answer
Using CyberChef, decode the file named "encodedflag.txt" and use regex to extract by MAC address. What is the extracted value?
08-2E-9A-4B-7F-61
Correct Answer

TASK 9 – Log Analysis Tools : Yara and Sigma

Answer the questions below

What languages does Sigma use?

YAML ✓ Correct Answer

What keyword is used to denote the "title" of a Sigma rule?

title ✓ Correct Answer

What keyword is used to denote the "name" of a rule in YARA?

rule ✓ Correct Answer

TASK 10 - Conclusion

Answer the questions below

Click and continue learning!

No answer needed ✓ Correct Answer

Result:

Thus the TryHackMe room “Intro to Log Analysis” was completed successfully.